Information Technology University, Lahore

Applied Machine Learning

BSCE - 23

Spring-2026

Assignment # 02

Issue Date: **08/04/2025**

Due Date **12/04/2025** **Total marks =100**

Instructions

1. Please review the University Plagiarism Policy.
2. Any submissions handed in late will not be accepted and will receive a zero score.
3. Assignment should be uploaded as PDF file with source file in Google Classroom.
4. The name of the file should be your Roll Number as; BSCEXXXX.
5. Please submit your own work only.
6. You have to submit a .py code file.

Develop a **fully connected neural network (ANN)** to classify data into two categories (binary classification).

Dataset:

- Breast Cancer Wisconsin Dataset
- Dataset Link

Tasks:

1. Load and Explore the Dataset

- Load dataset manually using Pandas

Check:

- shape
- missing values
- class distribution

Normalize features (0-1 or standardization)

Manually split dataset:

- 80% training
- 20% validation

Do NOT use train_test_split()

2. Build a Custom Neural Network

- Build your own ANN **from scratch** (no pre-trained models).
- Input layer = number of features
- At least **2 hidden layers**

Use:

- ReLU activation
- Dropout (at least once)

Output layer:

1 neuron + sigmoid

- Print the model summary to show the architecture.

3. **Train the Model**
    - Compile the model using the Adam optimizer and binary cross-entropy loss.
    - Train the model for 20 epochs using the training data.
    - Use the validation data to monitor performance during training.

4. **Evaluate the Model**
    - Evaluate the model on the validation set and report:
        - Accuracy
        - Confusion Matrix (write a custom method to compute TP, TN, FP, FN using y_true and y_pred)
        - Precision, Recall, F1-Score
        - Plot the Confusion Matrix using Matplotlib
5. **Report Your Findings**
    - Summarize your methodology, results, and key findings in a short report.
    - Include:
        - A description of your custom ANN architecture.
        - Training and validation accuracy/loss curves.
        - Challenges faced and potential improvements.

Note: There should be Report.pdf detailing your experience and highlighting any interesting results. Kindly don’t explain your code in the report, just explain the results. Your report should include your comments on the results of all the steps,